Fear alone influences the behaviour of a keystone native prey by an invader; a non-consumptive effect of the catfish invasion of Aotearoa New Zealand's freshwaters

Olivier V. Raven Calum MacNeil Finnbar Lee
Robin Holmes Ian A. Kusabs Francis J. Burdon
Deniz Özkundakci

2026-03-30

Non-consumptive effects (NCEs) or 'fear' responses to predators encompass costly antipredator behaviours, such as increasing refuge use in conjunction with decreasing foraging and feeding efficiency. Such NCEs could influence prey population dynamics and even community structure and function, if the prey themselves are 'keystone' species. In Aotearoa-New Zealand, the native freshwater crayfish kōura (*Paranephrops planifrons*) is a keystone species and an ecosystem engineer but is preyed upon by native longfin eels (*Anguilla dieffenbachii*) and invasive brown bullhead catfish (*Ameiurus nebulosus*). Using controlled aquarium experiments, we tested whether visual and chemical cues from each predator induced changes in kōura refuge-seeking and feeding behaviour. Neither predator treatment significantly affected refuge use or feeding rate. The absence of a significant response to either predator, including the co-evolved eel, may reflect limited prior predator experience in the source lakes where eel densities are low. Substantial individual variation in kōura behaviour further reduced statistical power to detect treatment effects, highlighting the importance of accounting for personality differences in future NCE studies.

Table of contents

1 Introduction **2**

2 Materials and Methods **3**
2.1 Animal collection and preparation . . . 3
2.2 Refuge experiment . . . 4
2.3 Feeding experiment . . . 5
2.4 Data analysis . . . 5

3 Results **6**
3.1 Refuge experiment . . . 6
3.2 Feeding experiment . . . 6

4 Discussion **7**
4.1 Refuge experiment . . . 7
4.2 Feeding experiment . . . 7
4.3 Conclusion . . . 7

Acknowledgements **7**

References **7**

Tables **7**

Figures **7**

Supplementary information **10**

1 Introduction

Biological invasions can have profound impacts on invaded ecosystems, particularly through the introduction of novel predators that prey upon native species lacking an evolutionary history with them (Doherty et al., 2016). Native prey are often naïve to these predators, making them especially vulnerable to direct predation (Gamradt & Kats, 1996; Smith et al., 2008). In addition to these consumptive effects (CEs), native prey may exhibit heightened fear responses to invasive predators, leading to the development of anti-predator behaviours such as increased refuge use and avoidance of open habitats (Kindlinger & Albins, 2017). These behavioural changes, termed non-consumptive effects (NCEs), can substantially influence population dynamics by reducing foraging activity, growth, and reproductive output (Pressier et al., 2005). In contrast, native predators generally elicit different responses because prey and predators have co-evolved, resulting in behavioural and physiological adaptations that stabilise predator–prey interactions within local ecosystems (Abrams, 2000).

Freshwater crayfish rely primarily on chemical cues (kairomones) rather than visual cues to detect and respond to predatory fish (Blake & Hart, 1993; Brönmark & Hansson, 2012). The kairomones carry signals via excreta, mucus, or skin, that crayfish use to assess predator

risk, responding more strongly to predators that have consumed conspecifics and to alarm cues released when prey are injured or killed (Gherardi et al., 2011; Beattie & Moore, 2018). Exposure to kairomones from predatory fish can generate NCEs, including reduced growth and reduced activity levels in crayfish (Adams, 2007; Ramberg-Pihl & Yurewicz, 2020), and even non-predatory fish can cause these behaviour changes, likely through heightened generalised awareness of risk (Nyström, 2005). However, the strength of these responses depends on prior experience as wild-caught individual crayfish show stronger antipredator responses than farmed conspecifics naive to predators (Martin, 2014), indicating that both individual experiences and evolutionary history shape kairomones detection.

In Aotearoa-New Zealand (hereafter Aotearoa), the endemic northern freshwater crayfish *Paranephrops planifrons* White, 1842 commonly known as kōura is a 'culturally treasured' taonga species and valued food resource for Māori the indigenous people of Aotearoa (Kusabs & Quinn, 2009). Ecologically, kōura play a critical role in freshwater ecosystems as ecosystem engineers through their effects on sediment dynamics, nutrient cycling, and benthic community structure, and have been described as keystone species in many freshwater systems (Collier at al., 1997; Parkyn et al., 2001; Usio & Townsend, 2004). Kōura have a long co-evolutionary history with the native longfin eel (*Anguilla dieffenbachii*) and shortfin eel (*A. australis*) (MacNeil et al., 2025) and have been shown to respond more strongly to eel kairomones than to those of introduced non-native brown trout (*Salmo trutta*), even when sourced from streams where trout are present (Shave et al., 1994). This suggests that co-evolutionary history, rather than individual experience alone, shapes kōura predator recognition (Shave et al., 1994). Currently kōura are threatened by multiple interacting stressors, including climate change, water-quality degradation and invasive species, contributing to widespread population declines (Lee et al., 2025; Kusabs et al., in revision). One of the invasive species that threatens the kōura is the non-native brown bullhead catfish (*Ameiurus nebulosus*; Barnes & Hicks, 2003; Dedual, 2019), which has been introduced in Aotearoa in 1878 and has spread over various waterways on the North Island (McDowall, 1990; Lee et al., 2025). The brown bullhead catfish is highly tolerant of a wide range of environmental conditions (Lee et al., 2025) and, due to its broad gape and opportunistic feeding behaviour, is capable of consuming a wide variety of native aquatic organisms (Scott & Crossman, 1973; Dedual, 2019). The recent and ongoing spread of brown bullhead catfish into North Island waterways therefore represents a novel predation threat to which kōura may lack adequate behavioural or sensory adaptations.

In this study, controlled aquarium experiments were used to test whether predator presence and associated chemical cues induce changes in refuge-seeking and feeding behaviour, providing insight into the behavioural mechanisms underlying NCEs of kōura responses to native and invasive predation threats. It is expected that kōura would show more shelter-seeking behaviour when presented with eels compared to catfish, due to naivety of the kōura towards catfish. Also, it is expected that light regime would have an influence on shelter seeking behaviour of this nocturnal species. For feeding behaviour it is expected that kōura will feed less when exposed to eel kairomones but as they will be naïve to catfish kairomones, feeding will be similar to the control. The findings of this study contribute to understanding the behavioural mechanisms

underlying kōura vulnerability to invasive catfish and provide a basis for predicting population-level impacts as catfish continue to spread into uninvaded waterways across Aotearoa.

2 Materials and Methods

2.1 Animal collection and preparation

On August 19th, 2025, kōura were sourced from two catfish-free lakes, (Ōkāreka and Tikitapu (Blue Lake)) in the Rotorua Te Awara Lakes area, Bay of Plenty region of the North Island. On September 25th, 2025, Brown bullhead catfish and longfin eels were sourced from Lake Rotoroa in Hamilton in the Waikato region of the North Island. Experiments were conducted at the Aquatic Research Centre of the University of Waikato. Some kōura were previously used in a mesocosm habitat preference study but never exposed to a predator. They were acclimated for a week in holding tanks under controlled conditions: a 10:14 light: dark cycle reflecting the correct seasonal day-length at a stable temperature of 15.6 °C, pH 8.4 and 10 mg l^{1} dissolved oxygen. Kōura were fed every other day fish food pellets (Hikari Tropical carnivore pellets, animal protein 47%). Catfish and eels were held in separate holding tanks, with the same environmental and feeding regime to kōura.

Because kōura are taonga (culturally important) species, permission had to be sought from relevant iwi (Māori tribal unit), as well as relevant government regulator to source a limited number of animals. Experiments were adapted, particularly the feeding study to make the best use of the finite number of collected animals. As this study focused on kōura the catfish and eels were reused repeatedly in the study.

2.2 Refuge experiment

Prior to the experiments, all animals were individually weighed and measured. Only male kōura were used. Kōura had a mean (± SD) orbital carapace length (OCL) of 26.0 mm (± 3.08) (18.35-32.95 mm range) and a mean wet weight of 11.7 g (± 4.25) (3–23 g range). Catfish ranged in length from 260-267 mm and 2.34 – 3.16 kg wet weight and eels ranged in length from ? and 2.50-3.52 kg wet weight.

Experiments were conducted in 38 L glass tanks (600 × 200 × 220 mm) with each tank having three sides (apart from front of tank) blacked out to block visual cues from neighbouring tanks. Each tank was divided into two compartments by a 0.5 mm steel mesh divider, allowing visual and chemical cues between compartments, while preventing physical contact between animals on either side of barrier (Figure 1). All tanks were filled with 26.4 L dechlorinated tap water and had identical environmental regimes as previously detailed for acclimatisation tanks, with water flow provided by an air stone lodged the top of the barrier in each tank. Each tank housed a single kōura on one side of the barrier and either a single predator (catfish or eel)

or no predator (control) on the other side. A halved terracotta pot (90 mm diameter × 90 mm depth) was positioned on the kōura side of the tank to serve as a potential refuge. To control for potential tank or side effects, the position of predator treatments within the aquaria was systematically rotated between experimental rounds. The experiment was repeated across five rounds, resulting in 11 catfish replicates, 11 eel replicates, and 10 control replicates. This experiment was first carried out in the light (n = 32 tanks) before being repeated in the dark (n = 32 tanks) with a total of 64 tanks used. Observations were made in the dark using red lighting positioned above the tanks in the same location as the previous normal lighting for the light regime observations. Following each individual experimental run, each tank was thoroughly cleaned and refilled with dechlorinated tap water before reuse. Each individual kōura was only exposed once to a fish predator and there were no mortalities of kōura or fish during the course of experiments. All experiments were conducted within a three-week period (from 1/10/2025 to 16/10/2025), the dark trials following on from the light trials.

Observations of each kōura location was made 21 times at the following time intervals (in minutes): 0, 5, 10, 20, 30, 45, 60, 61, 65, 70, 80, 90, 120, 150, 210, 240, 245, 260, 275, 290, and 300. An individual kōura were placed in the middle of the tank compartment containing the refuge at 0-minute (its initial location being immediately noted after its addition to the tank) and was removed after the final 300-minute observation. The fish predator was introduced after 60 minutes into the tank on the other side of the barrier from the kōura compartment and removed again at 240 minutes. The control tanks had no fish added but at 60 minutes and again at 240 minutes the water in the 'predator' compartment was disturbed by mixing with a small net for approximately 30 seconds. This was to simulate the disturbance time associated with adding and removing fish in the other treatments. Kōura location within each tank was recorded at each time interval as either refuge (the animal was using the refuge, being inside it or immediately behind it and touching it) or in the open - non-refuge (any other location in the tank where the animal was not in the refuge or touching the refuge).

2.3 Feeding experiment

The feeding behaviour experiments were executed in 16 glass tanks of 9.25L (320 × 170 × 170 mm) filled with 5 L dechlorinated tap water and a halved terracotta pot (90 mm diameter × 90 mm depth). In three rounds a total of 46 male and two female kōura were placed individually in each tank. Kōura had a mean (± SD) orbital carapace length (OCL) of 27.84 mm (± 4.72) (20.48-41.15 mm range) and a mean wet weight of 15.65 g (± 8.84) (6-44 g range). For each round kōura were starved for different periods (0, 3 and 4 days) prior to the feeding experiment to also investigate how starvation period influenced feeding motivation.

Kairomones from catfish and eels were collected by placing 4 fish (catfish or eel) in a 10 L bucket filled with water from the holding tanks for 15 minutes to concentrate the kairomones in the water. 1 L from the buckets was added to the tank with the kōura in it and for the controls 1 L of dechlorinated tap water was added. Next, 10 Hikari Tropical carnivore pellets were added to each tank. After 24 hours, when each experiment terminated, the number of

pellets eaten (to the nearest quarter pellet) was counted and the feeding rate (pellets consumed per g kōura) calculated. As stated previously, the feeding study was adapted to make the best use of the finite number of collected taonga animals and the following protocols reflect this. After 0 days starvation, 16 kōura (14 male and 2 female) were exposed to eel water (n = 5), catfish water (n = 6) or control water (n = 5). After 3 days starvation, 16 kōura (male only) were exposed to eel water (n = 6), catfish water (n = 5) or control water (n = 5). After 4 days starvation, 16 kōura (male only) were exposed to eel water (n = 5), catfish water (n = 5) or control water (n = 6).

2.4 Data analysis

To analyse kōura refuge use a beta-binomial generalized linear mixed-effects model (GLMM; BRON) fitted using the glmmTMB package in R (BRON; BRON). The response variable was the number of observations in refuge out of total observations per experiment round, treatment (catfish/eel/control), light condition (light, dark), period (before/during/after) included as fixed effects. A beta-binomial distribution was used to account for overdispersion in the binomial proportion data. Model selection was conducted by comparing candidate models differing in fixed effect structure and random effect inclusion using AIC, with the model with the lowest AIC selected as optimal. The need for a tank-level random intercept was evaluated by comparing AIC between models with and without this term. Model fit of the final model was assessed using DHARMa simulation-based residuals, including tests for overdispersion and zero-inflation (Hartig et al., 2024). Marginal means and pairwise contrasts between factor levels were obtained using the emmeans package (Lenth et al., 2026).

The feeding rate was analysed using a linear model with pellets consumed per gram body weight as the response, and treatment (catfish, eel, control) and starvation period (0, 3, 4) as fixed effects, including their interaction. Because each starvation period was conducted in a separate experimental round, starvation period and round were fully confounded by design, precluding the inclusion of round as a random effect. Overall significance of fixed effects was assessed using Type III ANOVA (car package; Fox & Weisberg, 2018), and model assumptions were verified using residual plots and testing normality of residuals using the Shapiro-Wilk test. Pairwise comparisons between treatment and starvation level combinations were conducted using estimated marginal.

3 Results

3.1 Refuge experiment

The proportion of time kōura spent in refuge varied across treatments, experimental periods, and light conditions (Figure 2, Table 1). Under dark conditions, refuge use increased across the experimental period in the catfish treatment, rising from 0.39 before to 0.7 after exposure, while

the control showed a smaller increase (0.37 to 0.47) and eel-exposed kōura showed little change (0.27 before, 0.41 during, and 0.36 after). Under light conditions, refuge use was generally higher than under dark conditions, with catfish (0.52 before to 0.73 after) and control (0.4 to 0.82) both showing increases across periods, while eel-exposed kōura showed a smaller increase (0.35 to 0.53).

A beta-binomial GLMM with experimental unit as a random intercept was the best supported model (lowest AIC). Full model coefficients are reported in Table 2. Refuge use was most affected by period with more kōura using the refuge as the experiment progressed. Refuge use was significantly more in the during period than in the before period ($p = 0.144$, Figure 3, Table 3). This was also the case between before and after period ($p = 0.007$), but not between during and after ($p = 0.409$). Neither treatment nor light condition significantly affected refuge use. No significant differences were found between catfish and eel ($p = 0.01$), catfish and control ($p =$), or control and eel ($p = 0.314$). Light condition was also not significant ($p =$).

3.2 Feeding experiment

The feeding experiment showed the highest mean (±SE) feeding rate was after 3 days of starvation in the catfish treatment (0.235 ± 0.037 pellets g ^{1}). The linear model showed starvation period had significant effect on feeding rate ($p = 4 \times 10^{-4}$, Table 4). Treatment had no significant effect on feeding rate ($p = 0.968$), and the interaction between the treatment and starvation period was not significant ($p = 0.082$), although a marginal trend was observed (Figure 4). No significant difference was found in feeding rate between catfish, eel, and control treatments within each of the starvation periods, this suggests that predator kairomones did not affect feeding behaviour in kōura within this experiment.

4 Discussion

4.1 Refuge experiment

4.2 Feeding experiment

4.3 Conclusion

Acknowledgements

References

Tables

Figures

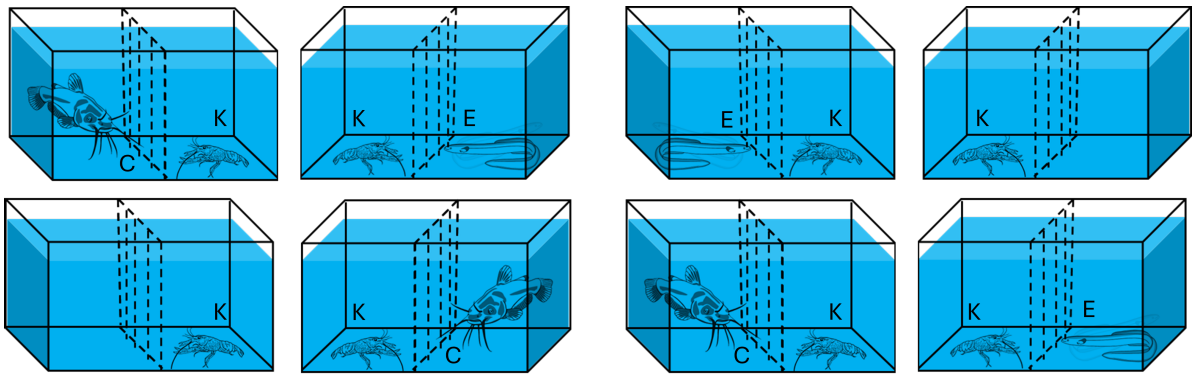

Figure 1: Schematic overview of the experimental aquarium layout for one experimental round. Eight aquaria were run simultaneously. In each tank, a single kōura (K) was housed on one side of the aquarium with a shelter, while a predator (catfish (C) or eel (E)) or no predator (control) was placed on the opposite side behind a mesh divider allowing visual and chemical cues but preventing physical contact.

`systemfonts and textshaping have been compiled with different versions of Freetype. Because c`

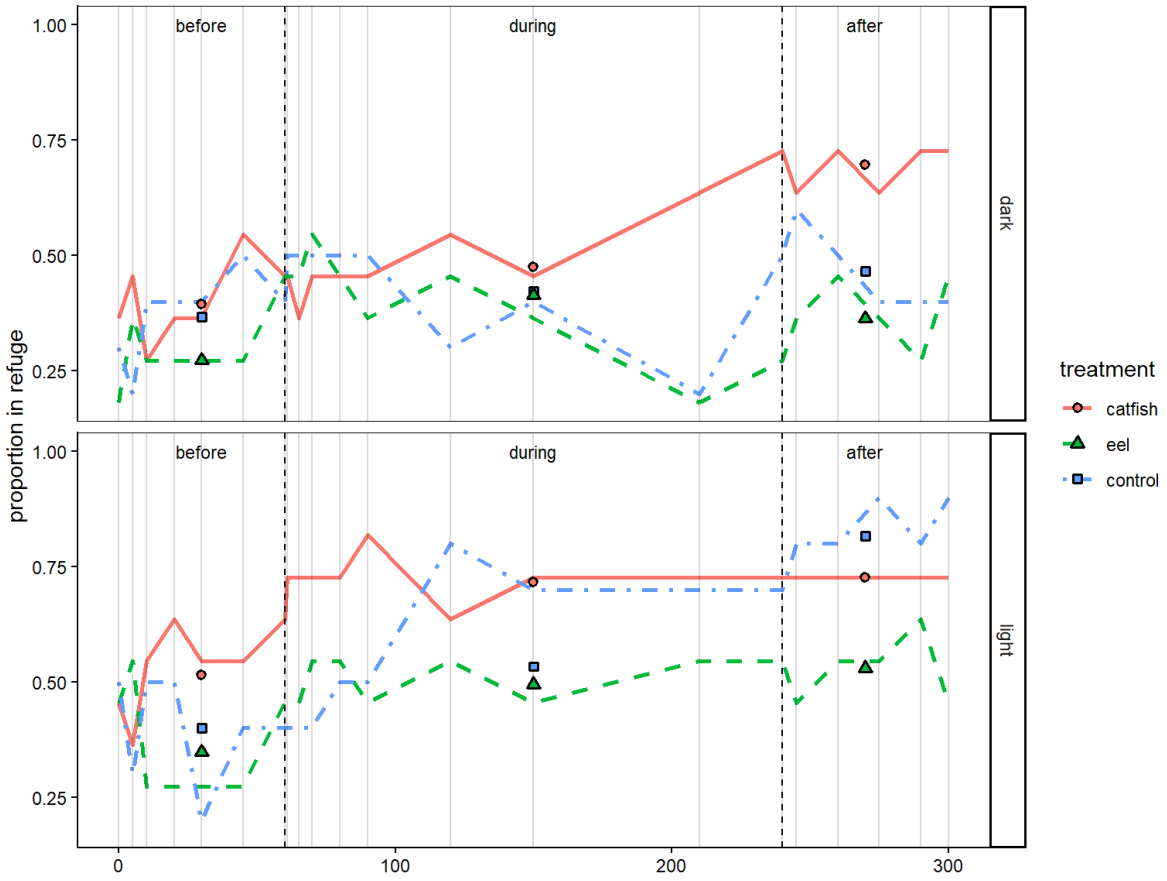

Figure 2: Kōura use of refuge over course of experiment with the three periods (before, during, after), the treatment (catfish; red-solid circle, eel; green-dashed triangle, control; blue-dotdash square), and divided over light regime (combined, dark, light). The points represent the proportion within each period for each treatment.

Source: Analysis Notebook

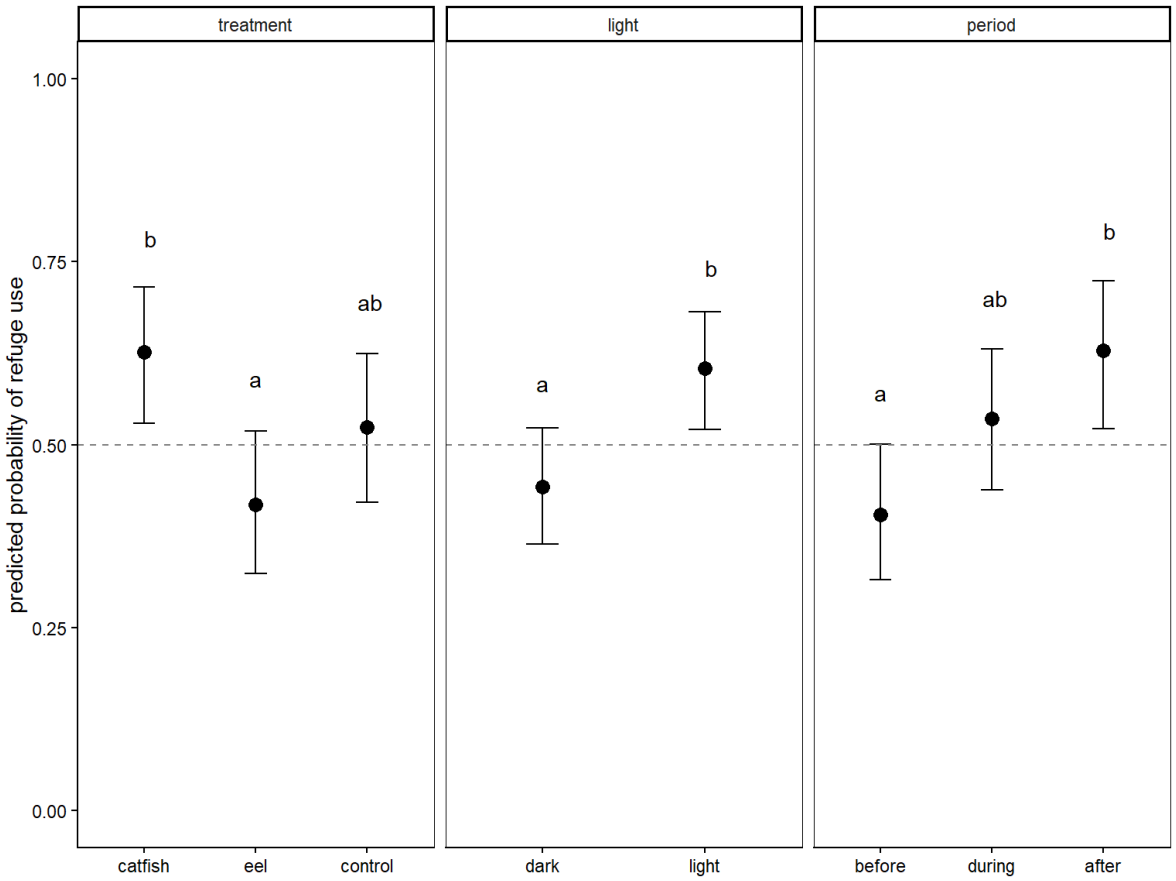

Figure 3: Predicted marginal probabilities of refuge use from the beta-binomial GLMM for each level of treatment, light condition, and experimental period. Points represent model-estimated marginal means averaged over other factors, with 95% confidence intervals shown as error bars. Different letters (a, b) indicate statistically significant differences between groups within each factor based on Tukey-adjusted pairwise comparisons. The dashed line at 0.5 marks the threshold where kōura spend half their time in refuge.

Source: Analysis Notebook

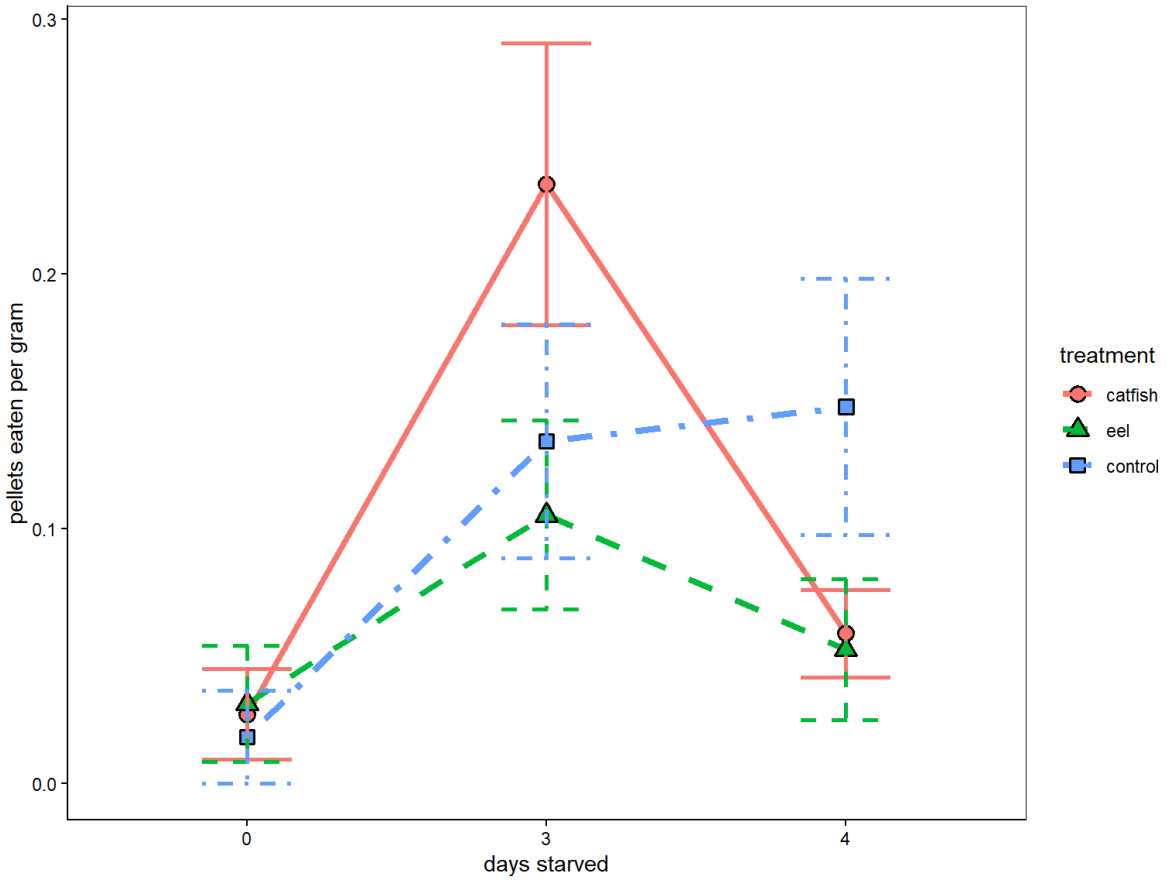

Figure 4: Mean kōura feeding rate (pellets consumed per gram body weight ± SE) across predator treatments (catfish, eel, control) and starvation periods (0, 3, and 4 days). Points represent treatment means and error bars show standard error.

Source: Analysis Notebook

Supplementary information

Source: Analysis Notebook

Table 2: Fixed effects of the beta-binomial GLMM for kōura refuge use.

Term	Estimate	Std. Error	z value	p value
(Intercept)	-0.058	0.282	-0.206	0.837
treatmentcatfish	0.424	0.292	1.452	0.146

Table 2: Fixed effects of the beta-binomial GLMM for kōura refuge use.

Term	Estimate	Std. Error	z value	p value
treatmenteel	-0.427	0.294	-1.452	0.147
lightdark	-0.654	0.239	-2.737	0.006
periodduring	0.530	0.282	1.881	0.060
periodafter	0.912	0.300	3.040	0.002

Source: Analysis Notebook

Table 3: Pairwise contrasts from the beta-binomial GLMM for kōura refuge use. Odds ratios and p-values are shown for treatment, light condition, and period comparisons. P-values are Tukey-adjusted for multiple comparisons within each factor.

Contrast	Odds ratio	SE	df	Null	z value	p value	Factor
control / catfish	0.655	0.191	Inf	1	-1.452	0.314	treatment
control / eel	1.533	0.451	Inf	1	1.452	0.314	treatment
catfish / eel	2.341	0.681	Inf	1	2.924	0.010	treatment
light / dark	1.924	0.460	Inf	1	2.737	0.006	light
before / during	0.589	0.166	Inf	1	-1.881	0.144	period
before / after	0.402	0.121	Inf	1	-3.040	0.007	period
during / after	0.682	0.205	Inf	1	-1.276	0.409	period

Source: Analysis Notebook

Table 4: Type III ANOVA table for the linear model of kōura feeding rate (pellets consumed per gram body weight). Treatment, starvation period, and their interaction were included as fixed effects.

Term	Sum Sq	Df	F value	p value
(Intercept)	0.004	1	0.640	0.429
treatment	0.000	2	0.032	0.968
day_starve	0.132	2	9.571	0.000
treatment:day_starve	0.062	4	2.241	0.082
Residuals	0.268	39	NA	NA

Source: Analysis Notebook

Table 1: Proportion of kōura observations recorded in refuge across experimental treatments (catfish, eel, control), light conditions (light, dark), and periods (before, during, and after predator exposure). Combined rows summarise across all levels of a given factor. Values represent the proportion of total observations where kōura were located in the refuge.

```
Unable to display output for mime type(s): text/html
```